Code No: 134BU **R16**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech II Year II Semester Examinations, April/May - 2023
OPERATING SYSTEMS
(Common to CSE, IT)

Time: 3 Hours **Max. Marks: 75**

Note: i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

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PART – A

(25 Marks)

1.a)	Draw the structure of OS.	[2]
b)	Write the objectives of operating system.	[3]
c)	Differentiate between process and thread.	[2]
d)	What is context switch?	[3]
e)	How the segmentation differ from paging?	[2]
f)	What is Virtual memory?	[3]
g)	What is File sharing?	[2]
h)	Describe the importance of seek( ).	[3]
i)	Which are the necessary conditions for Deadlock?	[2]
j)	What is language based protection?	[3]

PART – B

(50 Marks)

2.a) Explain about the operations of operating systems.
b) Explain how operating system used in a variety of computing environments. [5+5]

OR

3.a) Write the services provided by Operating System in terms of security and protection.
b) What is operating system? Explain the abstract view of the components of a computer system. [5+5]

4.a) What is monitor and give a solution for Reader's and writer's Critical section problem using monitors.
b) Explain about the process states with neat sketch. [5+5]

OR

5.a) What is Scheduler? Explain the queuing diagram representation of process scheduler with neat sketch.
b) Explain about the process control block with neat sketch. [5+5]

6. For the following page reference string 7,0,1,2,0,3,0,4,2,3,0,3,2,1,2,0,1,7,0,1. Calculate the page faults applying the following Page replacement algorithms for a memory with three frames:
   a) LRU
   b) FIFO. [5+5]

OR

7.a) What is Demand paging? Explain the working of demand paging and also give the advantages and disadvantages of Demand Paging.
   b) Explain the causes of Thrashing. [7+3]

8.a) Explain about FCFS disk scheduling algorithms with an example.
   b) Explain about the file system structure. [5+5]

OR

9.a) Explain about the various file access methods.
   b) Explain about free space management in file system. [5+5]

10.a) Explain the protection mechanism illustrating the use of protection domain and access control list.
   b) Explain the implementation of the Access Matrix. [5+5]

OR

11. Explain the methods for handling deadlocks. [10]

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